The Æther-flow Interpretation of Relativity

Primitive-Reservoir Observer-Reduction
Section-Centered Hidden-Response
Pre-Pre-Proto-Section Dynamics Collapse to an
Observer-Relevant Pullback-Quadratic Core

Abstract

The current active-primary theorem already removed the section-centered residual-normal-form datum $\mathfrak{R}^{\mathrm{sub,proto,presec,res}}$ as the primitive stopping point of the active pre-proto-section line by realizing the current benchmark-faithful pre-proto-section dynamics package through a deeper explicit section-centered hidden-response pre-pre-proto-section dynamics package $\mathfrak{S}^{\mathrm{sub,proto,pppresec}}$. That theorem renewed the live burden at hidden-response scope. The present note proves that a further honest internal sharpening is available strictly inside that package.

For each sector, the benchmark-facing content of the current hidden-response package collapses to the smaller observer-relevant pullback-quadratic core $\mathfrak{S}^{\mathrm{sub,proto,pppresec,pb}}$: the compact section-centered displacement body together with the positive pullback operator

$$A_\sigma^{\mathrm{pppresec,pb}}(y) = L_\sigma^{\mathrm{pppresec}}(y)^* K_\sigma^{\mathrm{pppresec}}(y) L_\sigma^{\mathrm{pppresec}}(y)$$

on the section-centered displacement space. The separate hidden-response space, hidden-response body, response-map factorization, and hidden operator no longer carry independent benchmark-facing freedom once that pullback core is fixed. Conversely, every such pullback core admits a canonical hidden-response realization with identity response map. Hence the current hidden-response frontier is not yet minimal: the live burden contracts from the larger hidden-response package to the smaller pullback-quadratic core.

The benchmark boundary remains fixed throughout. The one operative metric is still exactly $g^\sharp$, the ordinary matter route is unchanged, there is no second operative metric, the low-energy matter law is not enlarged, and no stronger promotion language is licensed. The positive result is therefore a genuine theorem-level collapse strictly inside the current hidden-response package. What remains open is to derive or justify the pullback-quadratic core from still more primitive explicit substrate structure, or else to prove a still sharper collapse, necessity result, or uniqueness result strictly inside that core.

1. Introduction

The active derivational program of Æther-flow Interpretation of Relativity again reaches a point where depth alone is not enough. The current active-primary theorem already showed that the current section-centered residual-normal-form datum is not primitive: it is induced by a deeper explicit section-centered hidden-response package $\mathfrak{S}^{\mathrm{sub,proto,pppresec}}$. That theorem mattered because it moved the live burden below the residual-normal-form presentation itself.

The present note asks the next disciplined question. Once the hidden-response package is treated as the current frontier, which parts of that package are genuinely independent benchmark-facing data, and which parts are only a factorized realization of a smaller datum already sufficient to recover the same current residual law and therefore the same current benchmark-faithful pre-proto-section dynamics? The answer proved here is that the benchmark-facing content of the current hidden-response package is already carried by a smaller pullback-quadratic core on the section-centered displacement fibers.

Framework and claim boundary. Æther / Æther-flow framework denotes the broader Æther / Æther-flow framework built on the claims that the Æther is the underlying four-dimensional substrate of reality, the Æther-flow is its intrinsic ordered motion, observed three-dimensional space is the local experiential slice of that deeper substrate, S-time is the experienced order of change arising from matter, light, and the Æther-flow, and observed expansion is the three-dimensional appearance of deeper four-dimensional ordered motion. Within that framework, Æther-flow Interpretation of Relativity denotes the exact-closure theory in which the effective gravitational dynamics are adopted to be exactly Einsteinian with universal matter coupling. In the active sequence, *adoption* means use of that established relativistic dynamics without claiming substrate derivation, while *derivation* is reserved for a first-principles recovery from explicit substrate variables.

The present note stays strictly inside that boundary. It does not introduce a second operative metric, it does not enlarge the low-energy matter law, and it does not reopen the already-spent larger pre-proto-section presentations as if they were again live burdens. Its narrower benchmark-facing role is to prove that the current hidden-response package contains a smaller observer-relevant core that now sets the honest next burden.

2. Fixed Inputs

Definition 1 (Recorded primitive continuation data)**.** *Fix the following continuation data from the active primitive-reservoir observer-reduction line.*

1. *For each sector label* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$*, a primitive forcing shell*

$$\mathcal{I}_\sigma^{\mathrm{for}}.$$

2. *The product shell*

$$\mathcal{S}_{\mathrm{prim}} \equiv \mathcal{I}_{\mathrm{car}}^{\mathrm{for}} \times \mathcal{I}_{\mathrm{cur}}^{\mathrm{for}} \times \mathcal{I}_{\mathrm{hom}}^{\mathrm{for}}.$$

3. *The recorded metric and ordinary-matter outputs*

$$\mathscr{G}_{\mathrm{out}} : \mathcal{S}_{\mathrm{prim}} \to \{\textit{observer metrics}\}, \qquad \mathscr{T}_{\mathrm{out}} : \mathcal{S}_{\mathrm{prim}} \times \Psi \to \{\textit{observer stress tensors}\},$$

satisfying

$$\mathscr{G}_{\mathrm{out}}(\mathbf{w}) = g^\sharp, \qquad \mathscr{T}_{\mathrm{out}}(\mathbf{w}, \psi) = T_{\mu\nu}^{\mathrm{matter}}[g^\sharp, \psi] \tag{1}$$

for every $\mathbf{w} \in \mathcal{S}_{\mathrm{prim}}$ *and every matter configuration* $\psi \in \Psi$*.*

Definition 2 (Recorded proto-section benchmark base and downstream chain)**.** *Fix* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$*. The active line already fixes the following proto-section benchmark base and downstream consequences:*

1. *the current proto-section benchmark base* $\mathfrak{Q}^{\mathrm{sub,proto,secgen}}$*, presented at current scope through*

$$\mathcal{Q}_\sigma^{\mathrm{psec}}, \qquad \mathcal{B}_\sigma^{\mathrm{psec}}, \qquad \mathscr{E}_\sigma^{\mathrm{psec}}, \qquad \mathcal{Z}_\sigma^{\mathrm{psec}};$$

2. *the current observer-relevant pre-proto-section minimizer-section core* $\mathfrak{R}^{\mathrm{sub,proto,presec,sec}}$*, the current section-centered residual normal form* $\mathfrak{R}^{\mathrm{sub,proto,presec,res}}$*, and the current benchmark-faithful pre-proto-section package* $\mathfrak{R}^{\mathrm{sub,proto,presec}}$ *as already recorded on the active line;*

3. *the previously recorded fact that, once a benchmark-faithful section-centered residual-normal-form realization over the fixed proto-section benchmark base is available, the current pre-proto-section package, the current minimizer-section core, and the previously recorded seed, root, foundation, ground, origin, precursor, lift, shell-restriction, split-core, selector-law, combined-response, ambient-package, trace-core, exact-shell-affine-nucleus, continuity-Hessian compressed core, and benchmark one-metric observer-package consequences on the active line follow unchanged;*

4. *benchmark faithfulness, meaning preservation of the benchmark outputs* (1)*, no second operative metric, no enlarged low-energy matter law, no change to the ordinary matter route, and no premature promotion from adoption to completed derivation.*

Definition 3 (Section-centered hidden-response pre-pre-proto-section dynamics)**.** *Fix* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$*. A* section-centered hidden-response pre-pre-proto-section dynamics package $\mathfrak{S}^{\mathrm{sub,proto,pppresec}}$ *for sector* σ *consists of the following data.*

1. *A finite-dimensional Euclidean section-centered displacement space*

$$V_\sigma^{\mathrm{pppresec}}$$

and a compact convex section-centered displacement body

$$\mathcal{D}_\sigma^{\mathrm{pppresec}} \subset \mathcal{B}_\sigma^{\mathrm{psec}} \times V_\sigma^{\mathrm{pppresec}}$$

with nonempty interior in each fiber

$$\mathcal{D}_\sigma^{\mathrm{pppresec}}(y) \equiv \{v \in V_\sigma^{\mathrm{pppresec}} : (y, v) \in \mathcal{D}_\sigma^{\mathrm{pppresec}}\}.$$

Each fiber is required to be nonempty, compact, and convex, and

$$(y, 0) \in \mathcal{D}_\sigma^{\mathrm{pppresec}} \qquad \textit{for every } y \in \mathcal{B}_\sigma^{\mathrm{psec}}.$$

2. *A finite-dimensional Euclidean hidden-response space*

$$H_\sigma^{\mathrm{pppresec}},$$

a compact convex hidden-response body

$$\mathcal{K}_\sigma^{\mathrm{pppresec}} \subset H_\sigma^{\mathrm{pppresec}}$$

with $0 \in \operatorname{int} \mathcal{K}_\sigma^{\mathrm{pppresec}}$*, and a smooth family of injective linear maps*

$$L_\sigma^{\mathrm{pppresec}}(y) : V_\sigma^{\mathrm{pppresec}} \to H_\sigma^{\mathrm{pppresec}}, \qquad y \in \mathcal{B}_\sigma^{\mathrm{psec}},$$

such that

$$L_\sigma^{\mathrm{pppresec}}(y)\big(\mathcal{D}_\sigma^{\mathrm{pppresec}}(y)\big) \subset \mathcal{K}_\sigma^{\mathrm{pppresec}}$$

for every $y \in \mathcal{B}_\sigma^{\mathrm{psec}}$*.*

3. *A smooth family of positive self-adjoint hidden operators*

$$K_\sigma^{\mathrm{pppresec}}(y) : H_\sigma^{\mathrm{pppresec}} \to H_\sigma^{\mathrm{pppresec}}, \qquad y \in \mathcal{B}_\sigma^{\mathrm{psec}},$$

satisfying the uniform coercive bound

$$\langle \eta, K_\sigma^{\mathrm{pppresec}}(y)\eta \rangle \geq \mu_\sigma \|\eta\|^2 \qquad \text{for all } y \in \mathcal{B}_\sigma^{\mathrm{psec}},\ \eta \in H_\sigma^{\mathrm{pppresec}} \tag{2}$$

for some constant $\mu_\sigma > 0$*.*

4. *Exact-shell discipline, meaning that the distinguished exact shell of the package is the zero displacement slice over the fixed proto-section shell:*

$$\{(y, 0) \in \mathcal{D}_\sigma^{\mathrm{pppresec}} : y \in \mathcal{Z}_\sigma^{\mathrm{psec}}\}.$$

5. *Benchmark faithfulness, meaning preservation of the benchmark outputs* (1)*, no second operative metric, no enlarged low-energy matter law, and presentation as a deeper explicit substrate-side source for the current section-centered residual datum rather than as a replacement benchmark theory.*

Define the induced residual functional

$$\widetilde{\mathscr{R}}_\sigma^{\mathrm{ppsec,res}} : \mathcal{D}_\sigma^{\mathrm{pppresec}} \to [0, \infty)$$

by

$$\widetilde{\mathscr{R}}_\sigma^{\mathrm{ppsec,res}}(y, v) \equiv \frac{1}{2} \langle L_\sigma^{\mathrm{pppresec}}(y)v, K_\sigma^{\mathrm{pppresec}}(y) L_\sigma^{\mathrm{pppresec}}(y)v \rangle. \tag{3}$$

Remark 1. *Definition 3 fixes the full current hidden-response package. The present note asks whether all of that package is benchmark-facingly necessary, or whether the active line already depends only on a smaller datum inside it.*

3. Observer-Relevant Pullback-Quadratic Core

Definition 4 (Observer-relevant pullback-quadratic core)**.** *Fix* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$ *and a benchmark-faithful hidden-response package in the sense of Definition 3. Define the pullback operator*

$$A_\sigma^{\mathrm{pppresec,pb}}(y) \equiv L_\sigma^{\mathrm{pppresec}}(y)^* K_\sigma^{\mathrm{pppresec}}(y) L_\sigma^{\mathrm{pppresec}}(y) : V_\sigma^{\mathrm{pppresec}} \to V_\sigma^{\mathrm{pppresec}}.$$

The observer-relevant pullback-quadratic core $\mathfrak{S}^{\mathrm{sub,proto,pppresec,pb}}$ *for sector* σ *consists of:*

1. *the section-centered displacement space* $V_\sigma^{\mathrm{pppresec}}$*;*

2. *the compact convex section-centered displacement body* $\mathcal{D}_\sigma^{\mathrm{ppprese c}} \subset \mathcal{B}_\sigma^{\mathrm{psec}} \times V_\sigma^{\mathrm{ppprese c}}$*;*

3. *the smooth family of positive self-adjoint pullback operators* $A_\sigma^{\mathrm{ppprese c,pb}}(y)$*;*

4. *the pullback residual functional*

$$\mathscr{P}_\sigma^{\mathrm{ppprese c,pb}} : \mathcal{D}_\sigma^{\mathrm{ppprese c}} \to [0,\infty), \qquad \mathscr{P}_\sigma^{\mathrm{ppprese c,pb}}(y,v) \equiv \frac{1}{2}\left\langle v, A_\sigma^{\mathrm{ppprese c,pb}}(y)v\right\rangle;$$

5. *the exact shell*

$$\{(y,0) \in \mathcal{D}_\sigma^{\mathrm{ppprese c}} : y \in \mathcal{Z}_\sigma^{\mathrm{psec}}\};$$

6. *benchmark faithfulness in the sense of Definition 2(4).*

From this core define the reconstructed section-centered pre-proto-section package by:

1. *the state space*

$$\widehat{\mathcal{R}}_\sigma^{\mathrm{ppsec}} \equiv \mathcal{Q}_\sigma^{\mathrm{psec}} \times V_\sigma^{\mathrm{ppprese c}};$$

2. *the body*

$$\widehat{\mathcal{B}}_\sigma^{\mathrm{ppsec}} \equiv \mathcal{D}_\sigma^{\mathrm{ppprese c}};$$

3. *the projection*

$$\widehat{\Pi}_\sigma^{\mathrm{ppsec}}(y,v) \equiv y;$$

4. *the functional*

$$\widehat{\mathscr{F}}_\sigma^{\mathrm{ppsec}}(y,v) \equiv \mathscr{E}_\sigma^{\mathrm{psec}}(y) + \mathscr{P}_\sigma^{\mathrm{ppprese c,pb}}(y,v);$$

5. *the exact shell*

$$\widehat{\mathcal{Z}}_\sigma^{\mathrm{ppsec}} \equiv \left\{(y,0) \in \widehat{\mathcal{B}}_\sigma^{\mathrm{ppsec}} : y \in \mathcal{Z}_\sigma^{\mathrm{psec}}\right\}.$$

Proposition 1 (Pullback-core properties)**.** *For each* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$*, the pullback-quadratic core of Definition 4 satisfies:*

1. $A_\sigma^{\mathrm{ppprese c,pb}}(y)$ *is self-adjoint and*

$$\left\langle v, A_\sigma^{\mathrm{ppprese c,pb}}(y)v\right\rangle \geq \mu_\sigma \left\|L_\sigma^{\mathrm{ppprese c}}(y)v\right\|^2 > 0$$

for every $y \in \mathcal{B}_\sigma^{\mathrm{psec}}$ *and every nonzero* $v \in V_\sigma^{\mathrm{ppprese c}}$*;*

2. $$\mathscr{P}_\sigma^{\mathrm{ppprese c,pb}}(y,v) = \widetilde{\mathscr{R}}_\sigma^{\mathrm{ppsec,res}}(y,v) \qquad \textit{for every } (y,v) \in \mathcal{D}_\sigma^{\mathrm{ppprese c}};$$

3. *for each fixed* $y \in \mathcal{B}_\sigma^{\mathrm{psec}}$*, the unique zero of*

$$v \mapsto \mathscr{P}_\sigma^{\mathrm{ppprese c,pb}}(y,v) \qquad \textit{on } \mathcal{D}_\sigma^{\mathrm{ppprese c}}(y)$$

is $v = 0$*.*

Proof. Item (1) follows because $A_\sigma^{\mathrm{ppprese c,pb}}(y)$ is the adjoint pullback of the positive self-adjoint operator $K_\sigma^{\mathrm{ppprese c}}(y)$. More explicitly,

$$\left\langle v, A_\sigma^{\mathrm{ppprese c,pb}}(y)v\right\rangle = \left\langle L_\sigma^{\mathrm{ppprese c}}(y)v, K_\sigma^{\mathrm{ppprese c}}(y)L_\sigma^{\mathrm{ppprese c}}(y)v\right\rangle,$$

which is nonnegative by positivity of $K_\sigma^{\mathrm{ppresec}}(y)$ and strictly positive for $v \neq 0$ because $L_\sigma^{\mathrm{ppresec}}(y)$ is injective and (2) holds.

Item (2) is exactly the displayed identity above together with Definitions 3 and 4. Item (3) follows from item (1), because the quadratic form vanishes if and only if $L_\sigma^{\mathrm{ppresec}}(y)v = 0$, which by injectivity is equivalent to $v = 0$. □

Proposition 2 (The reconstructed package is benchmark-faithful pre-proto-section dynamics)**.** *For each $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$, the reconstructed data of Definition 4 form a benchmark-faithful section-centered realization of the current proto-section-generating substrate pre-proto-section dynamics package. More precisely:*

1. *$\widehat{\mathcal{B}}_\sigma^{\mathrm{ppsec}} \subset \widehat{\mathcal{R}}_\sigma^{\mathrm{ppsec}}$ is compact and convex, and*
$$\widehat{\Pi}_\sigma^{\mathrm{ppsec}}(\widehat{\mathcal{B}}_\sigma^{\mathrm{ppsec}}) = \mathcal{B}_\sigma^{\mathrm{psec}};$$
2. *each fiber*
$$(\widehat{\Pi}_\sigma^{\mathrm{ppsec}})^{-1}(y) \cap \widehat{\mathcal{B}}_\sigma^{\mathrm{ppsec}} = \{y\} \times \mathcal{D}_\sigma^{\mathrm{ppresec}}(y)$$
is nonempty, compact, and convex;
3. *for each fixed $y \in \mathcal{B}_\sigma^{\mathrm{psec}}$, the restriction of $\widehat{\mathscr{F}}_\sigma^{\mathrm{ppsec}}$ to that fiber is strictly convex and has the unique minimizer $(y, 0)$;*
4. $$\widehat{\mathcal{Z}}_\sigma^{\mathrm{ppsec}} = \left\{(y, 0) \in \widehat{\mathcal{B}}_\sigma^{\mathrm{ppsec}} : y \in \mathcal{Z}_\sigma^{\mathrm{psec}}\right\},$$
and
$$\widehat{\Pi}_\sigma^{\mathrm{ppsec}}\big|_{\widehat{\mathcal{Z}}_\sigma^{\mathrm{ppsec}}} : \widehat{\mathcal{Z}}_\sigma^{\mathrm{ppsec}} \xrightarrow{\cong} \mathcal{Z}_\sigma^{\mathrm{psec}}$$
is a diffeomorphism;
5. *the package is benchmark faithful in the sense of Definition 2(4).*

Proof. Items (1) and (2) are inherited directly from Definition 3(1): the displacement body is compact and convex, projects onto $\mathcal{B}_\sigma^{\mathrm{psec}}$, and each fiber is nonempty, compact, and convex.

For item (3), fix $y \in \mathcal{B}_\sigma^{\mathrm{psec}}$. On the fiber $\{y\} \times \mathcal{D}_\sigma^{\mathrm{ppresec}}(y)$,
$$\widehat{\mathscr{F}}_\sigma^{\mathrm{ppsec}}(y, v) = \mathscr{E}_\sigma^{\mathrm{psec}}(y) + \frac{1}{2}\left\langle v, A_\sigma^{\mathrm{ppresec,pb}}(y)v\right\rangle.$$
The first term is constant on the fiber. By Proposition 1(1), the second term is a positive-definite quadratic form in v. Hence the fiber restriction is strictly convex, and by Proposition 1(3) its unique minimizer is $v = 0$.

Item (4) is exactly the shell discipline built into Definition 4. Item (5) is inherited from the benchmark faithfulness of the original hidden-response package: only the benchmark-relevant pullback quadratic form is retained, and the fixed outputs (1) are unchanged. □

4. Canonical Hidden-Response Realization of the Pullback Core

Definition 5 (Canonical hidden-response realization)**.** *Fix $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$ and an observer-relevant pullback-quadratic core in the sense of Definition 4. Define the canonical hidden-response*

realization by:

1. *hidden-response space*
$$H_\sigma^{\mathrm{pppresec,can}} \equiv V_\sigma^{\mathrm{pppresec}};$$

2. *canonical hidden radius*
$$\rho_\sigma^{\mathrm{pppresec}} \equiv 1 + \sup\left\{\|v\| : (y,v) \in \mathcal{D}_\sigma^{\mathrm{pppresec}}\right\};$$

3. *canonical hidden body*
$$\mathcal{K}_\sigma^{\mathrm{pppresec,can}} \equiv \left\{\eta \in V_\sigma^{\mathrm{pppresec}} : \|\eta\| \le \rho_\sigma^{\mathrm{pppresec}}\right\};$$

4. *canonical response map*
$$L_\sigma^{\mathrm{pppresec,can}}(y) \equiv \mathrm{id}_{V_\sigma^{\mathrm{pppresec}}};$$

5. *canonical hidden operator*
$$K_\sigma^{\mathrm{pppresec,can}}(y) \equiv A_\sigma^{\mathrm{pppresec,pb}}(y).$$

Proposition 3 (Canonical realization reproduces the pullback core)**.** *For each $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$, the canonical data of Definition 5 form a benchmark-faithful section-centered hidden-response package in the sense of Definition 3, and their induced residual functional is exactly the pullback functional:*
$$\frac{1}{2}\left\langle L_\sigma^{\mathrm{pppresec,can}}(y)v, K_\sigma^{\mathrm{pppresec,can}}(y) L_\sigma^{\mathrm{pppresec,can}}(y)v\right\rangle = \mathscr{P}_\sigma^{\mathrm{pppresec,pb}}(y,v).$$

Proof. Because $\mathcal{D}_\sigma^{\mathrm{pppresec}}$ is compact, the supremum defining $\rho_\sigma^{\mathrm{pppresec}}$ is finite. Hence $\mathcal{K}_\sigma^{\mathrm{pppresec,can}}$ is a compact convex body with 0 in its interior. By construction, every fiber $\mathcal{D}_\sigma^{\mathrm{pppresec}}(y)$ lies inside that body. The canonical response map is injective, and the canonical hidden operator is smooth, positive, and self-adjoint because the pullback operator has those properties by Proposition 1(1). The same exact-shell zero slice and the same benchmark-faithfulness clause are inherited unchanged from the pullback core.

Finally, because the canonical response map is the identity,
$$\frac{1}{2}\left\langle L_\sigma^{\mathrm{pppresec,can}}(y)v, K_\sigma^{\mathrm{pppresec,can}}(y) L_\sigma^{\mathrm{pppresec,can}}(y)v\right\rangle = \frac{1}{2}\left\langle v, A_\sigma^{\mathrm{pppresec,pb}}(y)v\right\rangle = \mathscr{P}_\sigma^{\mathrm{pppresec,pb}}(y,v).$$
So the canonical hidden-response realization reproduces exactly the same residual law. □

5. Main Theorem

Theorem 1 (Current hidden-response frontier collapses to the pullback-quadratic core)**.** *Fix the primitive continuation data of Definition 1, the proto-section benchmark base and downstream chain of Definition 2, and for each sector $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$ a benchmark-faithful section-centered hidden-response package in the sense of Definition 3. Then:*

1. *the current hidden-response package induces an observer-relevant pullback-quadratic core $\mathfrak{S}^{\mathrm{sub,proto,pppresec,pb}}$ and a benchmark-faithful section-centered residual-normal-form realization*

through Definition 4 and Proposition 2;

2. *every such pullback-quadratic core admits a canonical hidden-response realization by Proposition 3;*

3. *if two benchmark-faithful hidden-response packages induce the same pullback-quadratic core, then they induce the same section-centered residual-normal-form realization and therefore the same current benchmark-faithful pre-proto-section dynamics at current scope;*

4. *consequently the current hidden-response frontier is not benchmark-facingly primitive: its live benchmark-facing content collapses to the smaller observer-relevant pullback-quadratic core.*

Proof. Item (1) is the content of Definitions 4 and Proposition 2. Item (2) is Proposition 3. For item (3), if two hidden-response packages induce the same pullback-quadratic core, then by Definition 4 they define the same displacement body and the same pullback residual functional. Hence they define the same reconstructed section-centered residual-normal-form realization and the same reconstructed pre-proto-section package. The previously recorded section-centered normal-form necessity / uniqueness theorem then implies that these realizations are benchmark-facingly the same current pre-proto-section dynamics at current scope.

Item (4) is the routing consequence of item (3). Once the same current benchmark-faithful pre-proto-section dynamics depend only on the pullback-quadratic core, the extra hidden-response space, hidden body, response-map factorization, and hidden-operator factorization no longer carry independent benchmark-facing freedom. They are a realization of the pullback core, not additional live burden beyond it. □

Corollary 1 (Downstream consequences are unchanged)**.** *Once the observer-relevant pullback-quadratic core of Definition 4 is fixed, all currently recorded downstream benchmark-facing consequences on the active line follow unchanged: the current pre-proto-section package, the current minimizer-section core, the current section-centered residual normal form, and the previously recorded seed, root, foundation, ground, origin, precursor, lift, shell-restriction, split-core, selector-law, combined-response, ambient-package, trace-core, exact-shell-affine-nucleus, continuity-Hessian compressed core, and benchmark one-metric observer-package consequences remain exactly as before.*

Proof. Definition 2(3) records that these consequences already factor through a benchmark-faithful section-centered residual-normal-form realization over the fixed proto-section benchmark base. Proposition 2 supplies exactly such a realization from the pullback-quadratic core. Hence the downstream benchmark-facing consequences follow unchanged. □

Corollary 2 (The benchmark package remains fixed)**.** *Every benchmark-faithful pullback-quadratic core of Definition 4, every reconstructed package of Proposition 2, and every canonical hidden-response realization of Proposition 3 preserve the same benchmark one-metric package* (1)*. In particular, the operative metric remains exactly $g^\sharp$, the ordinary matter route is unchanged, there is no second operative metric, and the low-energy matter law is not enlarged.*

Proof. This is part of benchmark faithfulness in Definition 2(4). The present collapse only removes benchmark-neutral realization freedom inside the hidden-response package; it does not modify the fixed benchmark outputs. □

6. Routing Consequence

Corollary 3 (What the next honest move now is)**.** *After Theorem 1, the next honest benchmark-facing manuscript must either:*

1. *derive or justify the observer-relevant pullback-quadratic core itself from still more primitive explicit substrate structure in a way that changes benchmark-facing status;*
2. *identify a genuinely smaller theorem-level datum strictly inside that pullback core and prove a sharper collapse, necessity result, or uniqueness result there;*
3. *do either of the previous two while preserving the same benchmark one-metric package, the same ordinary-matter route, and the same claim boundary.*

Another same-output deeper-origin relay that merely preserves the same pullback-quadratic core and therefore merely reproduces the same current hidden-response package, the same current residual-normal-form datum, the same current pre-proto-section package, the same current minimizer-section core, the same proto-section benchmark base, and the same downstream benchmark-facing consequences, another reopening of the larger full hidden-response package, residual-normal-form presentation, or larger pre-proto-section presentations, or any move that introduces a second operative metric, enlarges the ordinary matter law, or uses stronger promotion language does not satisfy the present burden.

Proof. Theorem 1 shows that the current live hidden-response package carries no independent benchmark-facing content beyond the pullback-quadratic core. Therefore a subsequent manuscript must improve on that smaller core rather than merely restate the same current residual law or the same larger hidden-response package through another same-output relay. □

7. Conclusion

The positive result of this note is a genuine theorem-level collapse strictly inside the current hidden-response frontier. The current section-centered hidden-response pre-pre-proto-section dynamics package is not benchmark-facingly minimal once the fixed proto-section benchmark base is held fixed. Its benchmark-facing content is already carried by the smaller observer-relevant pullback-quadratic core on the section-centered displacement fibers.

This preserves the same benchmark one-metric package and the same claim boundary. The result does not yet provide a completed first-principles derivation of GR from final substrate microphysics, and it does not license stronger promotion language. Its narrower benchmark-facing gain is that the remaining honest burden is no longer the full current hidden-response package itself, but either a derivation or justification of the smaller pullback-quadratic core from still more primitive explicit substrate structure, or a still sharper collapse, necessity result, or uniqueness result strictly inside that core.